Ideal Secret Sharing Schemes over Small Domains

Amos Beimel - Aner Ben-Efraim - Oriol Farràs - **Adriana Moya**

Universitat Rovira i Virgili
Ben Gurion Univeristy
Ariel University

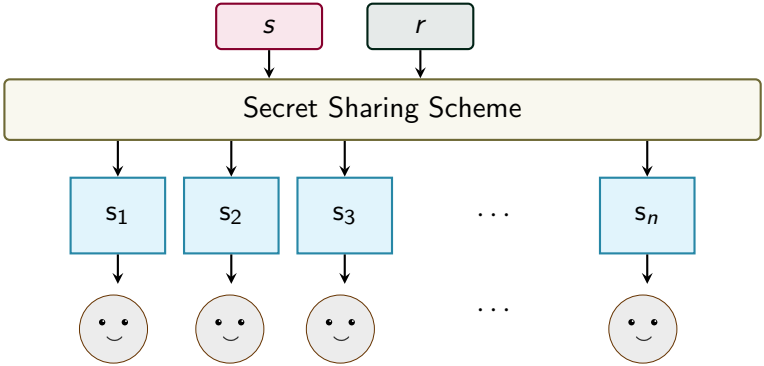
TCC 2026

A method to protect a secret

How? Distribute a secret s over a set of participants:

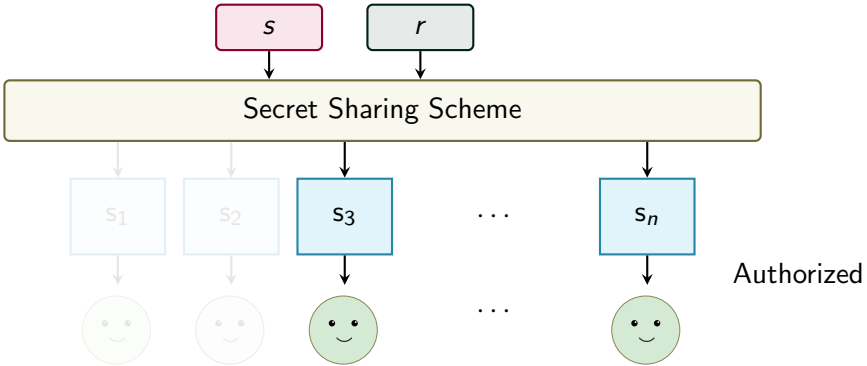
A method to protect a secret

How? Distribute a secret s over a set of participants:



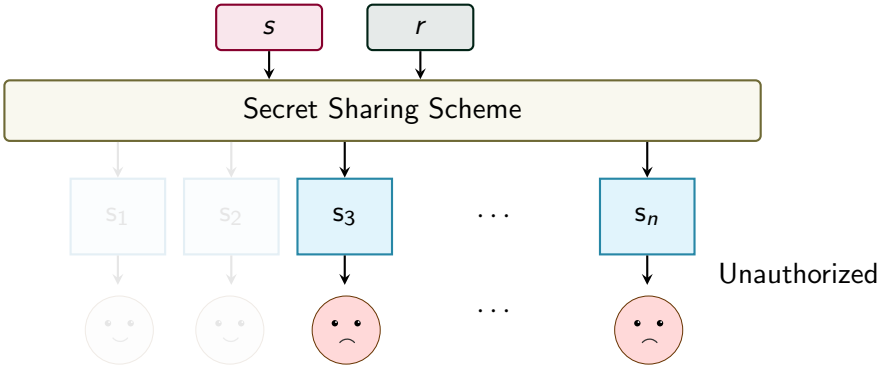
A method to protect a secret

How? Distribute a secret s over a set of participants:



A method to protect a secret

How? Distribute a secret s over a set of participants:



Access Structure

Access structure: $\Gamma \subseteq 2^P$ (Family of Authorized subsets)

Correctness: $A \in \Gamma$ Authorized sets can recover s

Privacy: $B \notin \Gamma$ Unauthorized sets cannot learn anything about s

$t = 2, n = 4$

Threshold Access Structure

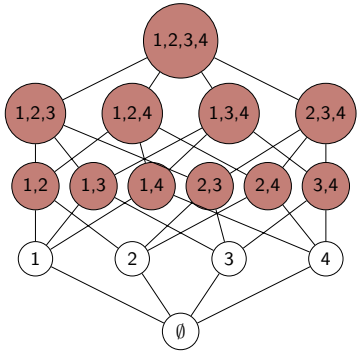
$\Gamma = \{A \subseteq \{1, \ldots, n\} : |A| \geq t\}$

Access Structure

Access structure: $\Gamma \subseteq 2^P$ (Family of Authorized subsets)

Correctness: $A \in \Gamma$ Authorized sets can recover s

Privacy: $B \notin \Gamma$ Unauthorized sets cannot learn anything about s



$t = 2, n = 4$

Threshold Access Structure

$$\Gamma = \{A \subseteq \{1, \ldots, n\} : |A| \geq t\}$$

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Secret Sharing Schemes

Properties:

- Grant access/privileges to sets of parties
- Prevent information leakage
- Reach agreements

Applications:

- Secure multiparty computation
- Threshold Cryptography
- Access Control
- Oblivious Transfer
- ...

Objective: Efficiency of schemes. Shares must be small

Ideal Secret Sharing Schemes

- Secret domain: $\mathcal{S}_0$
- Share domain: $\mathcal{S}_1, \ldots, \mathcal{S}_n$

Share size is always at least the secret size:

$$|\mathcal{S}_i| \geq |\mathcal{S}_0|$$

(Karnin, Greene, Hellman'83)

- $k \geq 2$, a secret sharing scheme $(S_0, S_1, \ldots, S_n)$ is k-ideal if

$$|\mathcal{S}_0| = |\mathcal{S}_1| = \cdots = |\mathcal{S}_n| = k.$$

- Its access structure Γ is called k-ideal

- Shamir'79 Blakley'79:

  For any prime power $k \geq n$, (t, n)-threshold access structures are k-ideal

Ideal Secret Sharing Schemes

- Secret domain: $\mathcal{S}_0$
- Share domain: $\mathcal{S}_1, \ldots, \mathcal{S}_n$

Share size is always at least the secret size:

$$|\mathcal{S}_i| \geq |\mathcal{S}_0|$$

(Karnin, Greene, Hellman'83)

- $k \geq 2$, a secret sharing scheme $(S_0, S_1, \ldots, S_n)$ is k-ideal if

$$|\mathcal{S}_0| = |\mathcal{S}_1| = \cdots = |\mathcal{S}_n| = k.$$

- Its access structure Γ is called k-ideal

- Shamir'79 Blakley'79:

  For any prime power $k \geq n$, (t, n)-threshold access structures are k-ideal

Ideal Secret Sharing Schemes

- Secret domain: $\mathcal{S}_0$
- Share domain: $\mathcal{S}_1, \ldots, \mathcal{S}_n$

Share size is always at least the secret size:

$$|\mathcal{S}_i| \geq |\mathcal{S}_0|$$

(Karnin, Greene, Hellman'83)

- $k \geq 2$, a secret sharing scheme $(S_0, S_1, \ldots, S_n)$ is k-ideal if

$$|\mathcal{S}_0| = |\mathcal{S}_1| = \cdots = |\mathcal{S}_n| = k.$$

- Its access structure Γ is called k-ideal

- Shamir'79 Blakley'79:

  For any prime power $k \geq n$, (t, n)-threshold access structures are k-ideal

About this talk

Goal:

Secret sharing schemes with small shares

Understand the limits of information theoretic schemes

Main Results:

- Characterization of k-ideal access structures
- Complete characterization for $k = 4, 6$

Techniques:

Combinatorics:	Matroids, Latin Squares, MOLS
Algebra:	Group Theory, Chinese Reminder Theorem
Information Theory:	Entropic representations

About this talk

Goal:

Secret sharing schemes with small shares

Understand the limits of information theoretic schemes

Main Results:

- Characterization of k-ideal access structures
- Complete characterization for $k = 4, 6$

Techniques:

Combinatorics:	Matroids, Latin Squares, MOLS
Algebra:	Group Theory, Chinese Reminder Theorem
Information Theory:	Entropic representations

About this talk

Goal:

Secret sharing schemes with small shares

Understand the limits of information theoretic schemes

Main Results:

- Characterization of k-ideal access structures
- Complete characterization for $k = 4, 6$

Techniques:

Combinatorics:	Matroids, Latin Squares, MOLS
Algebra:	Group Theory, Chinese Reminder Theorem
Information Theory:	Entropic representations

Example

Ideal scheme for the $(2,3)$-threshold access structure:

$$s, r \in \mathbb{F}_3 : \quad \begin{aligned} s_1 &= r \\ s_2 &= s + r \pmod 3 \\ s_3 &= s + 2r \pmod 3 \end{aligned} \quad \Longleftrightarrow \quad (s, r) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} = (s_1, s_2, s_3)$$

- Ideal
- Linear: Sharings are linear combination of the secret and the randomness

Example

Ideal scheme for the $(2,3)$-threshold access structure:

$$s, r \in \mathbb{F}_3 : \quad \begin{aligned} s_1 &= r \\ s_2 &= s + r \pmod 3 \\ s_3 &= s + 2r \pmod 3 \end{aligned} \quad \Longleftrightarrow \quad (s, r) \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} = (s_1, s_2, s_3)$$

- Ideal
- Linear: Sharings are linear combination of the secret and the randomness

Can we do better with non-linear?

- Linear schemes: Share size is $O(2^{0.5n})$ (tight) (Nir'26)
- General schemes:

  Share size is $O(2^{0.494n})$ (Nir'26)

  The best lower bound is $\Omega(n)$ (Csirmaz'97,Williamson'26)

Can we do better with non-linear?

- Linear schemes: Share size is $O(2^{0.5n})$ (tight) (Nir'26)
- General schemes:
  - Share size is $O(2^{0.494n})$ (Nir'26)
  - The best lower bound is $\Omega(n)$ (Csirmaz'97,Williamson'26)

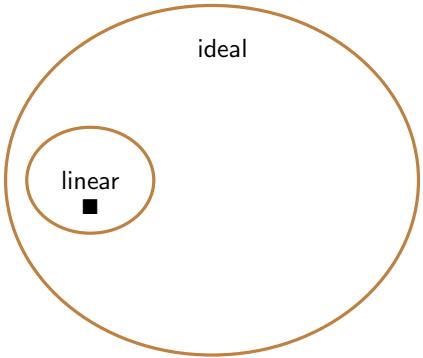
Can we do better with non-linear?

- Linear schemes: Share size is $O(2^{0.5n})$ (tight) (Nir'26)
- General schemes:

  Share size is $O(2^{0.494n})$ (Nir'26)

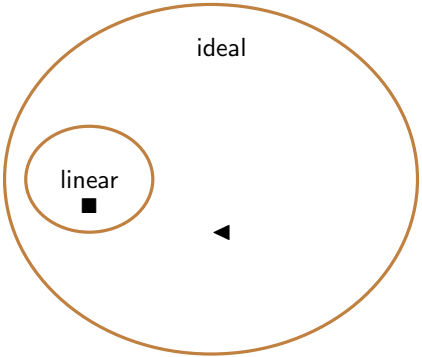
  The best lower bound is $\Omega(n)$ (Csirmaz'97,Williamson'26)

Focus on the ideal case



- Threshold Access Structure (Shamir'79 Blakley'79)

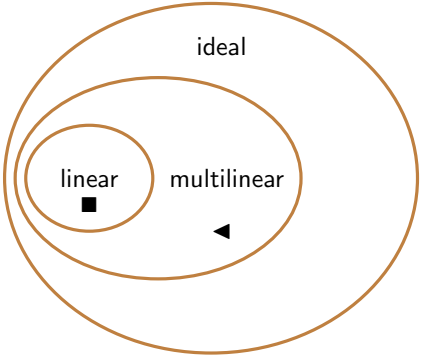
Focus on the ideal case



■ Threshold Access Structure (Shamir'79 Blakley'79)

◀ Non-Pappus Matroid: (Simonis Ashikmin'98)

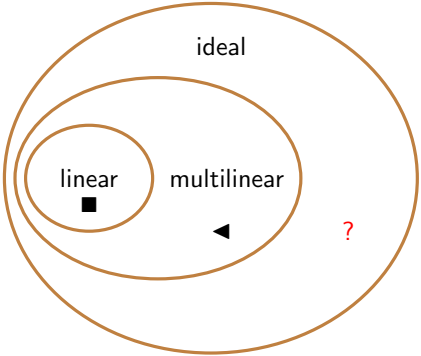
Focus on the ideal case



■ Threshold Access Structure (Shamir'79 Blakley'79)

◀ Non-Pappus Matroid: (Simonis Ashikmin'98)

Focus on the ideal case



■ Threshold Access Structure (Shamir'79 Blakley'79)

◀ Non-Pappus Matroid: (Simonis Ashikmin'98)

? Are there ideal access structures with no multilinear scheme?

Actual Characterization of k-Ideal Schemes

Known Results:

2-ideal	$\iff$	Ideal $\mathbb{F}_2$-linear
3-ideal	$\iff$	Ideal $\mathbb{F}_3$-linear
2 and 3-ideal	$\iff$	Ideal $\mathbb{F}_q$-linear $\forall$ prime power $q \geq 2$

Lehman'64, Seymour'76,
Beimel Chor'92, Matúš'93, Golic'98

This work:

4-ideal	$\iff$	Ideal $\mathbb{F}_4$-linear
6-ideal	$\iff$	Ideal $\mathbb{F}_2$-linear and $\mathbb{F}_3$-linear

Complete characterization by minors
New techniques

Actual Characterization of k-Ideal Schemes

Known Results:

2-ideal	$\iff$	Ideal $\mathbb{F}_2$-linear
3-ideal	$\iff$	Ideal $\mathbb{F}_3$-linear
2 and 3-ideal	$\iff$	Ideal $\mathbb{F}_q$-linear $\forall$ prime power $q \geq 2$

Lehman'64, Seymour'76,
Beimel Chor'92, Matúš'93, Golic'98

This work:

4-ideal	$\iff$	Ideal $\mathbb{F}_4$-linear
6-ideal	$\iff$	Ideal $\mathbb{F}_2$-linear and $\mathbb{F}_3$-linear

Complete characterization by minors
New techniques

- Matroid: Combinatorial structure $\mathcal{M} = (\{0, 1, \ldots, n\}, r)$
  Generalization of linear (in)dependence

Ideal Secret Sharing Scheme	$\Longrightarrow$	Matroid	Brickell Davenport'89
Ideal Linear Scheme	$\Longleftrightarrow$	Linear Matroid	

- Matroid: Combinatorial structure $\mathcal{M} = (\{0, 1, \ldots, n\}, r)$
  Generalization of linear (in)dependence

Ideal Secret Sharing Scheme	$\Longrightarrow$	Matroid	Brickell Davenport'89
Ideal Linear Scheme	$\Longleftrightarrow$	Linear Matroid	

Theorem (Geelen, Gerards, Kapoor'00)

The following are equivalent for Γ*:*

1. Γ *admits an ideal* $\mathbb{F}_4$*-linear scheme*
2. Γ *is determined by a matroid with no minor* $U_{2,6}, U_{4,6}, P_6, F_7^-, \left(F_7^-\right)^*, P_8, P_8''$

THEOREM

The following are equivalent for an access structure Γ*:*

1. Γ *admits an ideal* $\mathbb{F}_4$*-linear secret sharing scheme*
2. Γ *is determined by a matroid with no minor* $U_{2,6}, U_{4,6}, P_6, F_7^-, \left(F_7^-\right)^*, P_8, P_8''$
3. Γ *is 4-ideal (our result)*

Proof:

- Complete characterization of $\mathbb{F}_4$-linear matroids the forbidden minors (Geelen Gerards Kapoor'00)
- Every access structure with no $\mathbb{F}_4$-linear ideal scheme is not 4-ideal.

THEOREM

The following are equivalent for an access structure Γ*:*

1. Γ *admits an ideal* $\mathbb{F}_4$*-linear secret sharing scheme*
2. Γ *is determined by a matroid with no minor* $U_{2,6}, U_{4,6}, P_6, F_7^-, \left(F_7^-\right)^*, P_8, P_8''$
3. Γ *is 4-ideal (our result)**

Proof:

- Complete characterization of $\mathbb{F}_4$-linear matroids the forbidden minors (Geelen Gerards Kapoor'00)
- Every access structure with no $\mathbb{F}_4$-linear ideal scheme is not 4-ideal.

* This result was also proved by Kalantaria Khazaei'26 by using different techniques.

THEOREM

The following are equivalent for an access structure Γ*:*

1. Γ *admits ideal* $\mathbb{F}_2$*-linear and* $\mathbb{F}_3$*-linear schemes*
2. Γ *is* k*-ideal for every* $k \geq 2$
3. Γ *is determined by a matroid with no minor* $U_{2,4}, F_7, (F_7)^*$

Theorem

The following are equivalent for an access structure Γ:

1. Γ *admits ideal $\mathbb{F}_2$-linear and $\mathbb{F}_3$-linear schemes*
2. Γ *is k-ideal for every $k \geq 2$*
3. Γ *is determined by a matroid with no minor $U_{2,4}, F_7, (F_7)^*$*
4. Γ *is 6-ideal (our result)*

THEOREM

The following are equivalent for an access structure Γ*:*

1. Γ *admits ideal* $\mathbb{F}_2$*-linear and* $\mathbb{F}_3$*-linear schemes*
2. Γ *is* k*-ideal for every* $k \geq 2$
3. Γ *is determined by a matroid with no minor* $U_{2,4}, F_7, (F_7)^*$
4. Γ *is 6-ideal (our result)*

Proof:

- Complete characterization of access structures with $\mathbb{F}_2$ and $\mathbb{F}_3$-linear ideal schemes (Lehman'64, Seymour'76, Matúš'93, Beimel Chor'94, Golic'98)

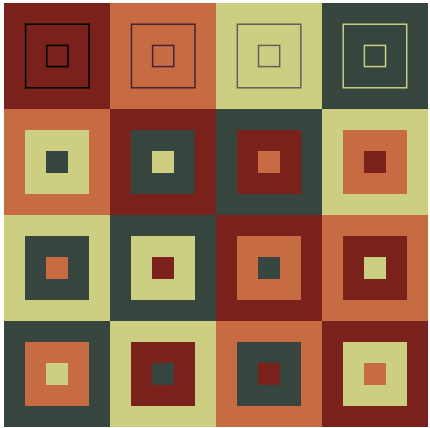
- Every access structure with no $\mathbb{F}_2$ nor $\mathbb{F}_3$-linear ideal schemes is not 6-ideal.

Techniques

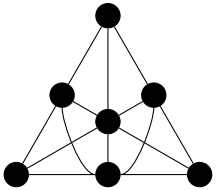
- Mutually Orthogonal Latin Hypercubes
- Group Based Latin Squares
- Group Characterizable Entropic Representations of Matroids

Open problems

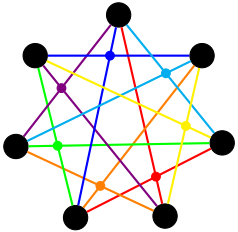
- Power of non-linear secret sharing schemes.
- Does every ideal access structure admit an ideal multilinear scheme?
- Characterization of k-ideal access structures for $k \geq 5$.
- Entropic Rota's Conjecture: For every k, is there a finite forbidden minor characterization for k-ideal schemes?



Thank you!



The Fano Matroid F_7

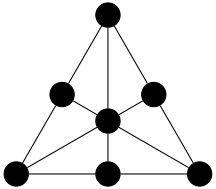


The Fano dual Matroid F_7^*

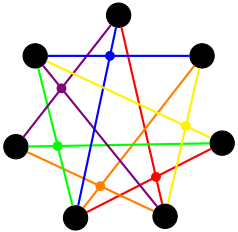
F_7 is k-entropic if and only if $k = 2^\ell$ for some $\ell > 0$. (Matúš'99)

Theorem

F_7^* is k entropic iff $k = 2^\ell$ for some $\ell > 0$.



The Non-Fano Matroid F_7^-



The Non-Fano dual Matroid $(F_7^-)^*$

F_7^- is k-entropic if and only if k is odd. (Matúš'99)

Theorem

$(F_7^-)^*$ is k-entropic iff k is odd.

Formal Statments

Theorem

If a (t, n)-threshold access structure has a k-ideal scheme then:

- $n \leq k + t - 2$ *for all* k *and all* $t \geq 2$.
- $n \leq k + t - 3$ *for* k *is odd and* $t \geq 3$.